CNN Math — Practice Questions

Based on: convolution and output size, parameter counting, convolution as matrix multiplication, max pooling, layer-by-layer output/weight/memory calculation, and normalization. No answers included.

1. The Convolution Operation and Output Size

Input image (5×5) and filter (3×3) for Q1 and Q2:

```
Input image:              Filter:
 2  0  1  3  1             1  0 -1
 1  2  0  1  2             1  0 -1
 0  1  3  2  0             1  0 -1
 2  1  0  1  1
 1  0  2  1  3
```

Q1. Using stride = 2 and padding = 1, find the size of the output feature map produced when the filter above is applied to the image above.

Q2. Using the same image, filter, stride, and padding as in Q1, find the output feature map.

Q3. An input of size 96×96 is convolved with a 7×7 filter, stride 2, and padding 0. Find the output size.

2. Counting Parameters in a Conv Layer

Q1. A convolutional layer takes a 128×128×3 input, uses 32 filters of size 5×5, stride 1, and padding 0. Find the output size, the total number of parameters, and the total number of neurons in the output.

3. Convolution as Matrix Multiplication

Q1. An input image of size 64×64×3 is convolved with 5×5 filters, stride 1, no padding, using 20 filters. Express this convolution as a matrix multiplication: give the dimensions of the image matrix, the filter matrix, and the resulting output matrix.

4. Max Pooling

Q1. Apply 2×2 max pooling with stride 2 to the following 4×4 feature map. Show the four 2×2 blocks you used and the resulting 2×2 output.

```
12   4   7   9
 3  15   2   6
 8   1  11   0
 5   9   3  14
```

Q2. A 6×6 feature map is pooled using 2×2 max pooling with stride 2. What is the size of the resulting output? What would the output size be instead if 3×3 pooling with stride 3 were used?

5. Layer-by-Layer Output Size, Weight, and Memory Calculation

CONV layers use 3×3 filters, stride 1, padding 1. POOL layers use 2×2 filters, stride 2.

Q1. Starting from an input of 64×64×3, compute the output size, memory, and number of weights for each of the following layers in order: (a) CONV3-32, (b) CONV3-32, (c) POOL2, (d) CONV3-64, (e) POOL2.

Q2. If a final fully-connected (FC) layer takes the flattened output of the last layer in Q1 and maps it to 5 output classes, how many weights does that FC layer alone contain?

6. Normalization Math

Formula: $x_norm = (x - \mu) / \sigma$

Image 1, Channel 1

3	5
7	9

Image 1, Channel 2

11	13
15	17

Image 2, Channel 1

19	21
23	25

Image 2, Channel 2

27	29
31	33

Q1. The boxes above show a batch of N = 2 images, each with C = 2 channels and a 2×2 spatial size. Normalize this data using Batch Norm, then using Layer Norm, then using Instance Norm. For each of the three, state clearly which boxed values you grouped together before normalizing.